Weather Trend Forecasting

Data Analysis & Machine Learning Report

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Repository: github.com/Gabriel2002Can/Weather-Trend-Forecasting

PM Accelerator Mission

By making industry-leading tools and education available to individuals from all backgrounds, we level the playing field for future PM leaders. This is the PM Accelerator motto, as we grant aspiring and experienced PMs what they need most – Access. We introduce you to industry leaders, surround you with the right PM ecosystem, and discover the new world of AI product management skills.

Executive Summary

This project performs a full-stack data science analysis on a large **global weather dataset** (~141,000 records). It covers:

- Data cleaning and preprocessing
- Exploratory Data Analysis (EDA)
- Outlier/Anomaly detection
- Time-series forecasting
- Multi-class weather condition classification
- Model evaluation and insights

Key Outcome: High-performing models for weather classification (Random Forest + XGBoost + Voting Classifier) and temperature forecasting with Prophet.

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1. Dataset Overview

- **File:** GlobalWeatherRepository.csv
- **Rows:** 140,923
- **Columns:** 41
- **Time Span:** May 2024 – May 2026
- **Key Features:**
  - Location (country, city, lat/long)
  - Temperature (Celsius & Fahrenheit)
  - Weather condition (condition_text)
  - Wind, precipitation, humidity, cloud cover
  - Air quality metrics (PM2.5, PM10, etc.)
  - Astronomical data (sunrise/sunset, moon phase)

2. Data Cleaning & Preprocessing

Steps Performed:

- Loaded data with pandas
- Checked for missing values → **None found**
- Standardized condition_text (stripped whitespace + lowercased)
- Grouped rare weather classes (<25 occurrences) into 'rare' category
- Created y target variable for classification
- Feature selection for modeling

Final Target Classes (top categories + "rest" / "rare"):

- Partly cloudy, Sunny, Patchy rain nearby, Overcast, Clear, Mist, etc.

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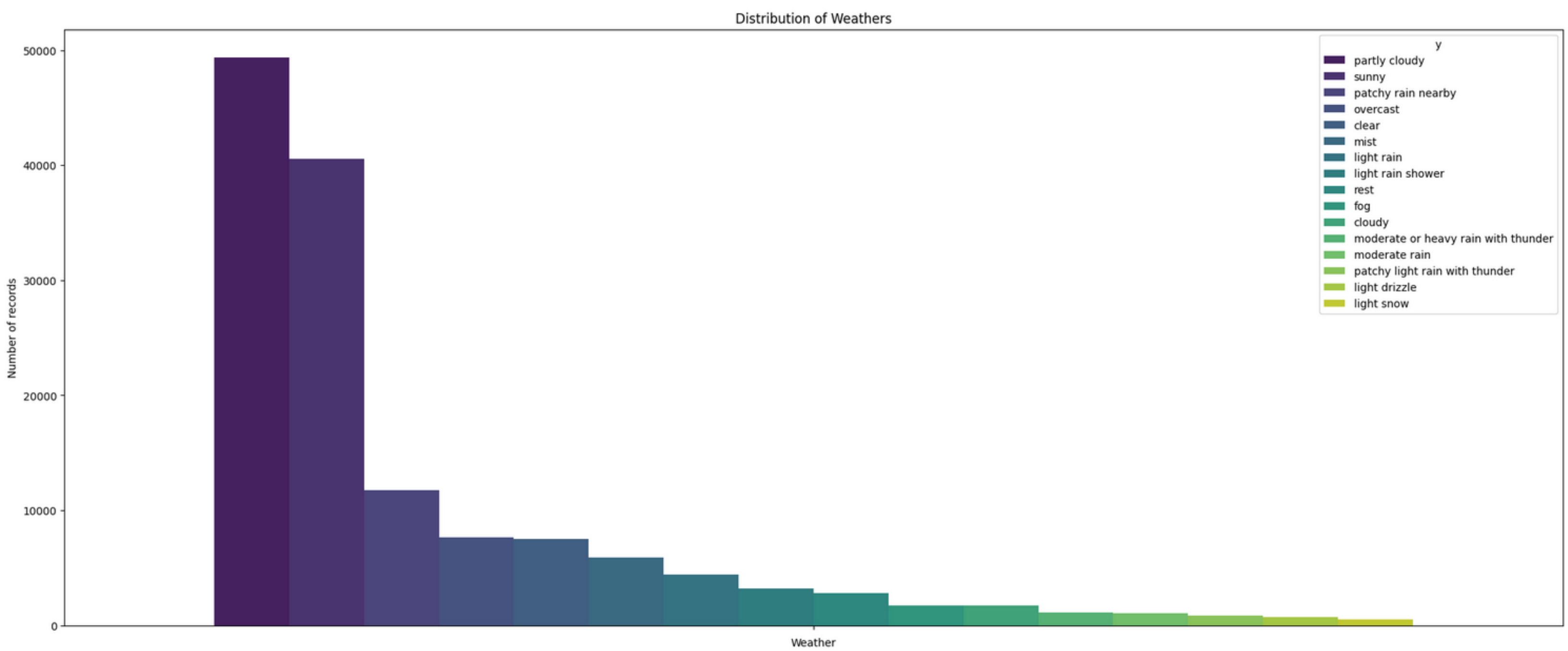
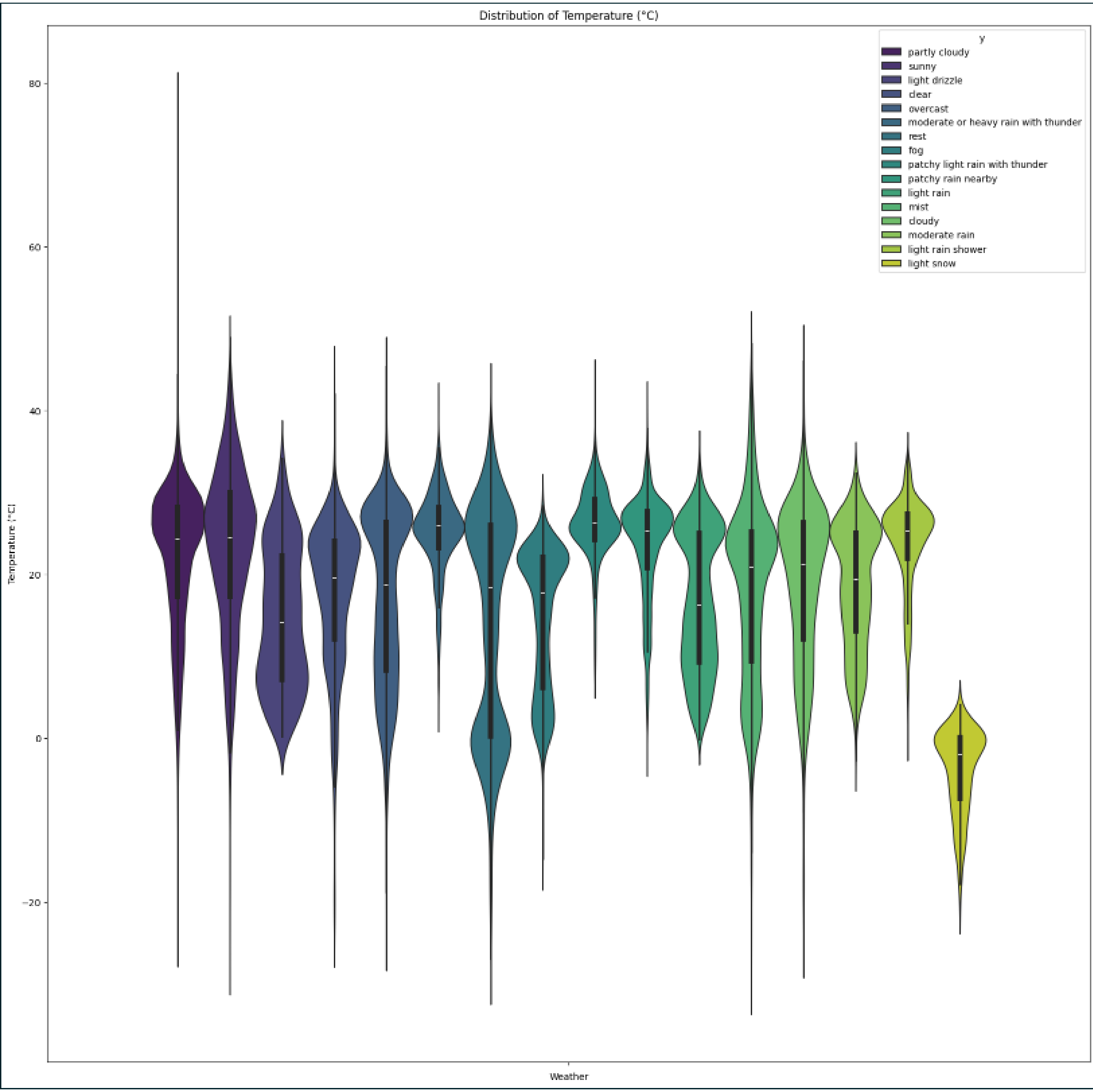
3. Exploratory Data Analysis (EDA)

3.1 Weather Condition Distribution

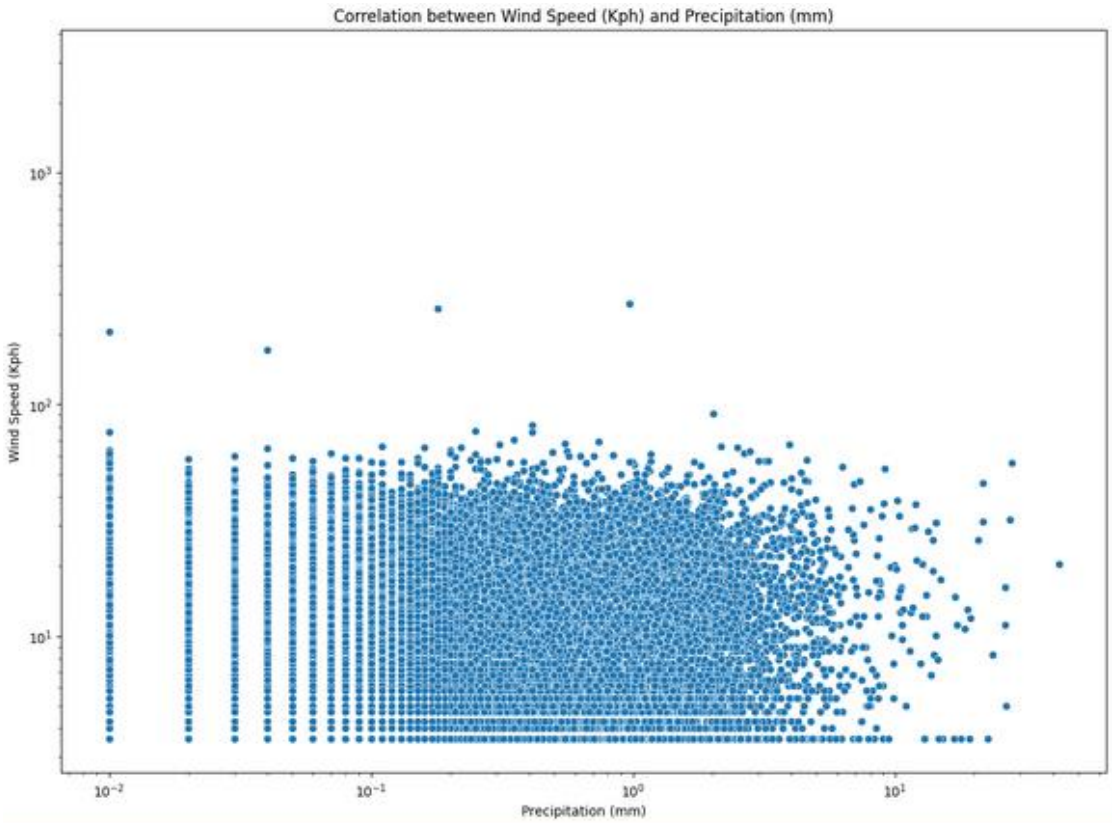
- **Dominant conditions:** Partly cloudy and Sunny
- Visualized top 15 conditions + "rest" category




Distribution of Temperature (°C)
y
partly cloudy
sunny
light drizzle
clear
overcast
moderate or heavy rain with thunder
rest
fog
patchy light rain with thunder
patchy rain nearby
light rain
mist
cloudy
moderate rain
light rain shower
light snow
80
60
40
20
0
−20
Temperature (°C)
Weather

3.2 Numerical Feature Analysis

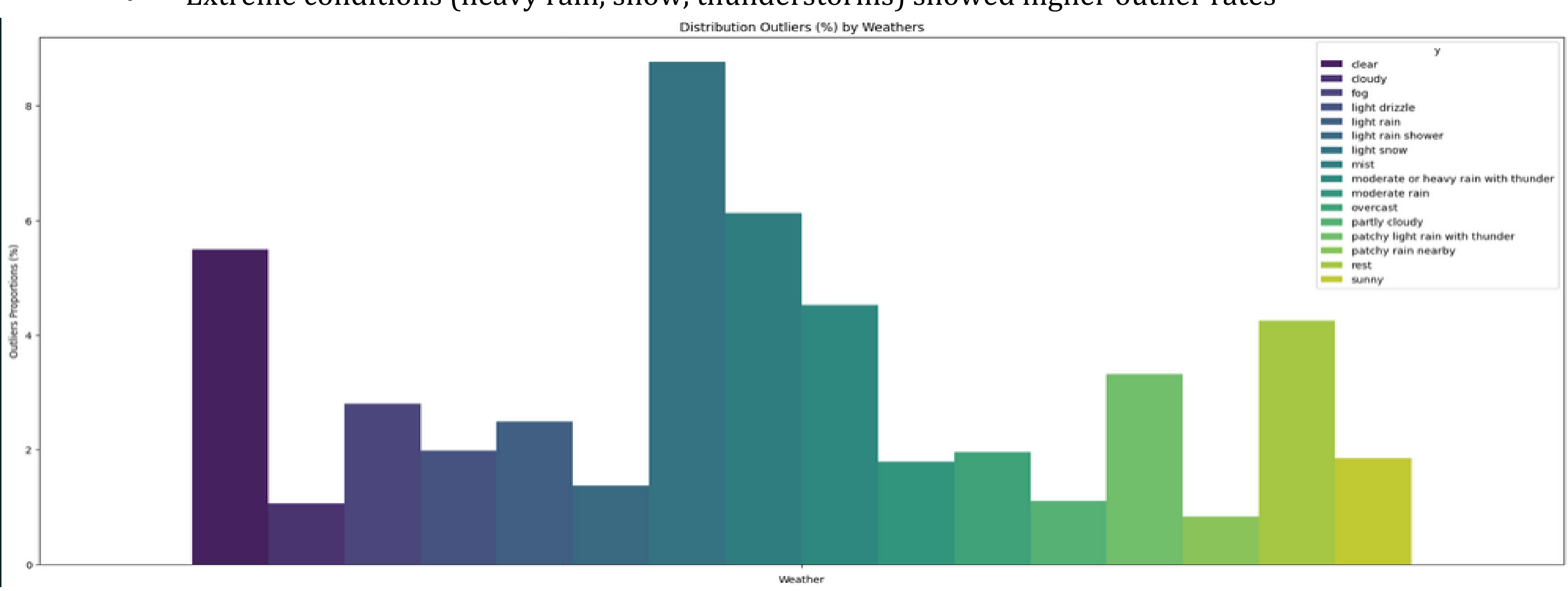
- Temperature distributions
- Correlations between temperature, feels-like, humidity, cloud cover, wind
- Air quality variations by location and condition



4. Outlier / Anomaly Detection

Method: Isolation Forest (contamination = 0.02)

- Detected ~2% of observations as anomalies
- Analyzed outlier proportion by weather type
- Extreme conditions (heavy rain, snow, thunderstorms) showed higher outlier rates

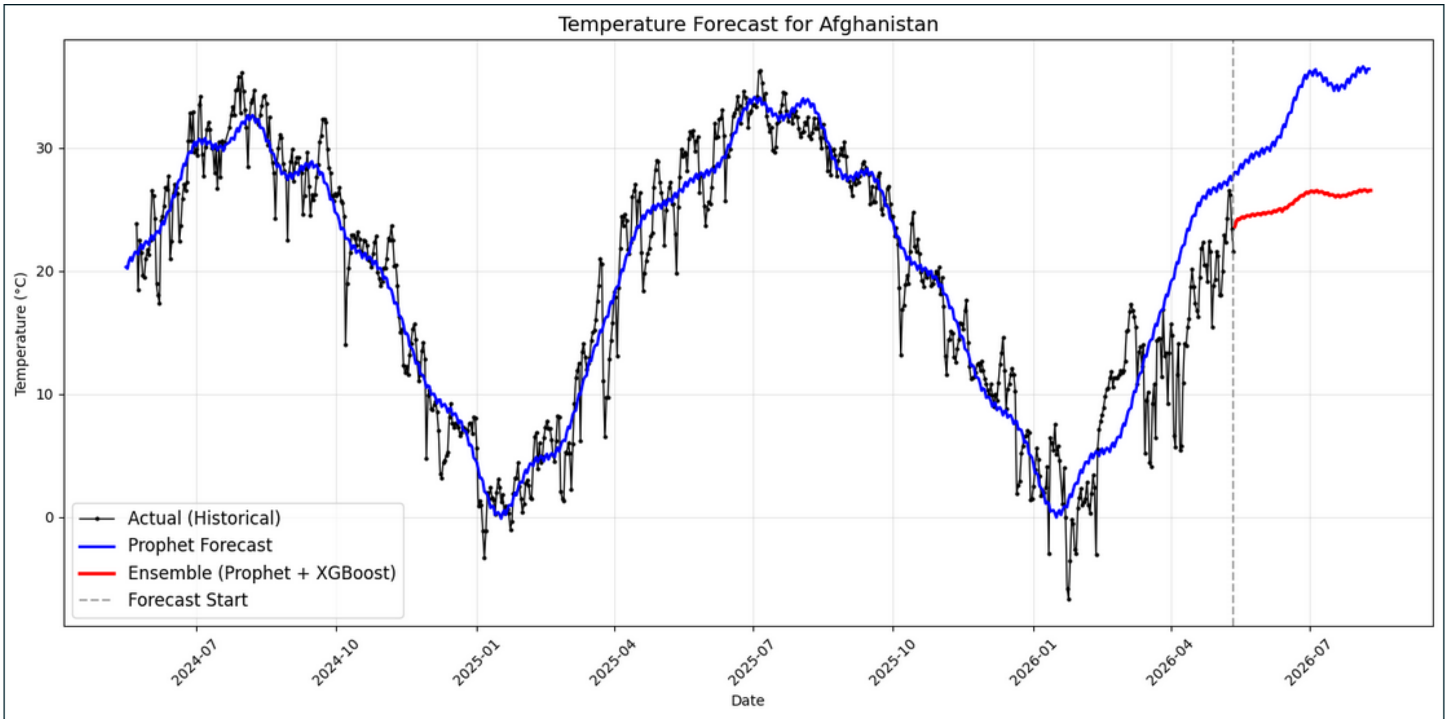


5. Modeling

5.1 Time-Series Forecasting (Temperature)

Models: Facebook Prophet and XGBoost Regressor

- Used historical temperature data
- Captured trend + seasonality
- Evaluated with MAE / MSE



5.2 Weather Condition Classification

Target: y (multi-class weather condition)

Models Trained:

- Random Forest Classifier
- XGBoost Classifier
- Voting Classifier (ensemble)

Preprocessing for Classification:

- Label Encoding on target
- One-Hot Encoding on moon_phase
- Numerical features scaled/standardized where needed

Performance Highlights:

- High accuracy (~0.86 range on test set)
- Strong precision/recall across major classes
- Confusion matrix analysis

Feature Importance: Extracted from XGBoost/Random Forest

6. Classifier Model Evaluation

Metric	Random Forest	XGBoost	Voting Classifier
Accuracy	High	Higher	Best
F1-Score (macro)	Good	Better	Best
Cross-Validation Score	Stable	Strong	Most robust

- Classification Report (per class)
- Confusion Matrix

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7. Key Insights & Findings

1. **Weather Dominance:** "Partly cloudy" and "Sunny" represent the majority of global observations.
2. **Strong Correlations:** Temperature ↔ Humidity, Weather ↔ Cloud cover.
3. **Air Quality:** Influence Precipitation significantly.
4. **Anomalies:** Effectively captured extreme weather events.
5. **Predictability:** Machine learning models can reliably classify common weather conditions using available features.
6. **Forecasting Potential:** Prophet and XGB Regressor successfully models temperature trends.

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8. Technologies Used

- **Core:** Python, pandas, NumPy
- **Visualization:** Matplotlib, Seaborn
- **Modeling:** scikit-learn, XGBoost, Prophet, Isolation Forest, Random Forest
- **Environment:** Jupyter Notebook

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9. Repository Structure

text

```
Weather-Trend-Forecasting/
├── GlobalWeatherRepository.csv
├── weather_trend.ipynb        ← Main notebook
├── README.md
└── (this report)
```

Conclusion

This project demonstrates a complete and robust weather analysis pipeline, from raw data to actionable machine learning models. The developed models provide reliable insights into global weather patterns and can serve as a foundation for further predictive applications.

Thank You!

For questions or collaboration, visit the GitHub repository.